Hide menu

Pauli X-Gate in Real IBM Quantum Computer

Video: Pauli X-Gate in Real IBM Quantum Computer
6 min

Graded Assignment:
Assessment 4
Submitted

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item

Assessment 4

1. Which Qiskit class is primarily used to create a quantum circuit? 1 / 1 point

   - QuantumRegister
   - Qubit
   - QuantumCircuit
   - ClassicalRegister

   Correct
   Correct! The QuantumCircuit class is used to create quantum circuits in Qiskit.

2. How can one visualize a quantum circuit in Qiskit? 1 / 1 point

   - Using the draw() method on the QuantumCircuit object
   - Using the visualize() method on the QuantumCircuit object
   - Using the plot() method on the QuantumCircuit object
   - Using the circuit_diagram() method on the QuantumCircuit object

   Correct
   Correct! The draw() method can be used to visualize a quantum circuit in Qiskit.

   Like Dislike Report an issue

3. What are the characteristics of the Pauli X gate in quantum computing? 1 / 1 point

   - [ ] It creates entanglement between qubits
   - [x] It can be represented by the Pauli-X matrix

   Correct
   Correct! The Pauli X gate is also known as the Pauli-X gate and can be represented by the Pauli-X matrix.

   - [ ] It applies a phase shift to the qubit
   - [x] It flips the state of a qubit

   Correct
   Correct! The Pauli X gate flips the state of a qubit from |0⟩ to |1⟩ and vice versa.

   - [x] It is a single-qubit gate

   Correct
   Correct! The Pauli X gate operates on a single qubit.

Review Learning Objectives

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

Assignment details

Due Dec 16, 11:59 PM IST
Attempts Unlimited
Submitted Dec 16, 1:31 AM IST
Your grade
100%
To pass you need at least 80%. We keep your highest score.
View submission See feedback
Retry

4. Which tool in Qiskit is primarily used to visualize the transition of quantum states in a circuit? 1 / 1 point

   - Qiskit Visualization
   - Qiskit Ignis
   - Qiskit Aer
   - Qiskit Terra

   Correct
   Correct! Qiskit Visualization provides tools for visualizing quantum states, circuits, and more.

5. In IBM Quantum Experience, what is the purpose of the 'Job Queue'? 1 / 1 point

   - To save quantum circuit designs
   - To debug quantum circuits
   - To manage and monitor submitted quantum circuit jobs
   - To provide tutorials and documentation

   Correct
   Correct! The Job Queue allows you to manage and monitor the status of your submitted quantum circuit jobs.